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Introduction to terminology

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DENT1020 Dental Science I

Learning outcomes

By the end of this lecture you should be able to:

Recognise the context of the study of anatomy and physiology

Correctly use anatomical terms to describe position and direction

Identify and describe the boundaries and associated surface landmarks of the oral cavity on a diagram and a patient

Suggested additional reading:
Fehrenbach MJ & Popowics T. Illustrated Dental Embryology, Histology, and Anatomy

Definitions

Anatomy

study of structure (*ana*- ‘up’, *tomia*- ‘cutting’)
macroscopic and microscopic

Physiology

study of function (*physis*- ‘nature’, *logia*- ‘study of’)

Anatomy and physiology are related disciplines

Correlation between structure and function is a core theme of biology (the s

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A Call to Action for Bioengineers and Dental Profe… Directives for the Future of TMJ Bioengineer…

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(Received 6 November 2006; accepted 13 March 2007; published online 29 March…

Abstract—The world's first TMJ Bioengineering Con… was held May 25–27, 2006, in Broomfield, C… Presentations were given by 34 invited speakers repr… industry, academics, government agencies such as N… private practice, which included surgeons, engine… medical scientists, and patient advocacy leader… attendees included documentary film makers a… officials. The impetus for the conference was tha… of TMJ research has been lacking continuity, wit… forum available for surgeons, scientists, and bioe… exchange scientific and clinical ideas and identi… goals, strengths, and capabilities. The goal was t… the seeds for establishing a forum for multidisc… interdisciplinary interactions. The collective … interactions brought about by a melting pot of… individuals has been pooled and is dissemi… article, which offers specific directives to bioen… scientists, and medical and dental professio… oral and maxillofacial surgeons, pain speciali… tists, prosthodontists, endocrinologists, rh… immunologists, radiologists, neurologists, a… surgeons. A primary goal of this article … researchers across a breadth of research ar… expertise to a significant clinical problem wit… new talent. For example, researchers with … element modeling will find an extensive … significant problems. Specific suggestions … were presented by the leading organi… surgeons and TMJ patients, and further r… identified in a series of group discussions… identified at the conference and prese…

RESEARCH REPORTS
Biomaterials & Bioengineering

Y. Lin[1], G.O. Gallucci[2], D. Buser[3], D. Bosshardt[3], U.C. Belser[4], and P.C. Yelick[1]*

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Bioengineered Periodontal Tissue Formed on Titanium Dental Implants

…e the durability and function of titanium dental implants … implant shape and design (Chen *et al.*, 2005; Lee *et al.*, … topography to control cell behavior (Oates *et al.*, 2007); …ce coatings (Catledge *et al.*, 2002; Wang *et al.*, 2005; …006; Hedia, 2007); and the use of growth factors (Park … osseointegration is currently considered as the optimal …e (Moradian-Oldak *et al.*, 2006), naturally formed teeth …nding alveolar bone *via* soft periodontal ligament (PDL) …erived progenitor cells have been shown to self-renew, …ple lineages, and function in periodontal tissue regenera… 2006; Kaneda *et al.*, 2006; Nagatomo *et al.*, 2006), their …gineering has emerged as a promising approach for the …ntal disease. A critical requirement for successful PDL …uccessful delivery of PDL progenitor cell populations …ng and differentiating *in situ*, in the oral environment. …old materials have been used in PDL tissue engineering, …ts for appropriate biocompatibility, biodegradability, cell …tor delivery, and mechanical stability (Abukawa *et al.*, … 2007). Here we describe the use of Matrigel (BD … Lakes, NJ, USA), a basement membrane matrix exten… cell differentiation (Kleinman and Martin, 2005), to … tissue formation on titanium implants. We predict that

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PLoS one

Functional Tooth Regeneration Using a Bioengineered Tooth Unit as a Mature Organ Replacement Regenerative Therapy

Masamitsu Oshima[1,9], Mitsumasa Mizuno[1,2,9], Aya Imamura[3], Miho Ogawa[1,4], Masato Yasukawa[3], Hiromichi Yamazaki[3], Ritsuko Morita[1], Etsuko Ikeda[2], Kazuhisa Nakao[1], Teruko Takano-Yamamoto[2], Shohei Kasugai[5], Masahiro Saito[1,3], Takashi Tsuji[1,3,4]*

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Abstract

Donor organ transplantation is currently an essential therapeutic approach to the replacement of a dysfunctional organ as a result of disease, injury or aging *in vivo*. Recent progress in the area of regenerative therapy has the potential to lead to bioengineered mature organ replacement in the future. In this proof of concept study, we here report a further development in this regard in which a bioengineered tooth unit comprising mature tooth, periodontal ligament and alveolar bone, was successfully transplanted into a properly-sized bony hole in the alveolar bone through bone integration by recipient bone remodeling in a murine transplantation model system. The bioengineered tooth unit restored enough the alveolar bone in a vertical direction into an extensive bone defect of murine lower jaw. Engrafted bioengineered tooth displayed physiological tooth functions such as mastication, periodontal ligament function for bone remodeling and responsiveness to noxious stimulations. This study thus represents a substantial advance and demonstrates the real potential for bioengineered mature organ replacement as a next generation regenerative therapy.

Citation: Oshima M, Mizuno M, Imamura A, Ogawa M, Yasukawa M, et al. (2011) Functional Tooth Regeneration Using a Bioengineered Tooth Unit as a Mature Organ Replacement Regenerative Therapy. PLoS ONE 6(7): e21531. doi:10.1371/journal.pone.0021531

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Competing Interests: The authors have declared that no competing interests exist.

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9 These authors contributed equally to this work.

Introduction

Donor organ transplantation is currently essential to replace a …

For the regeneration of ectodermal organs such as a tooth, hair follicle or salivary gland [16,17], a further concept has been proposed in which a bioengineered organ is developed from …

Levels of structural organisation

Chemical

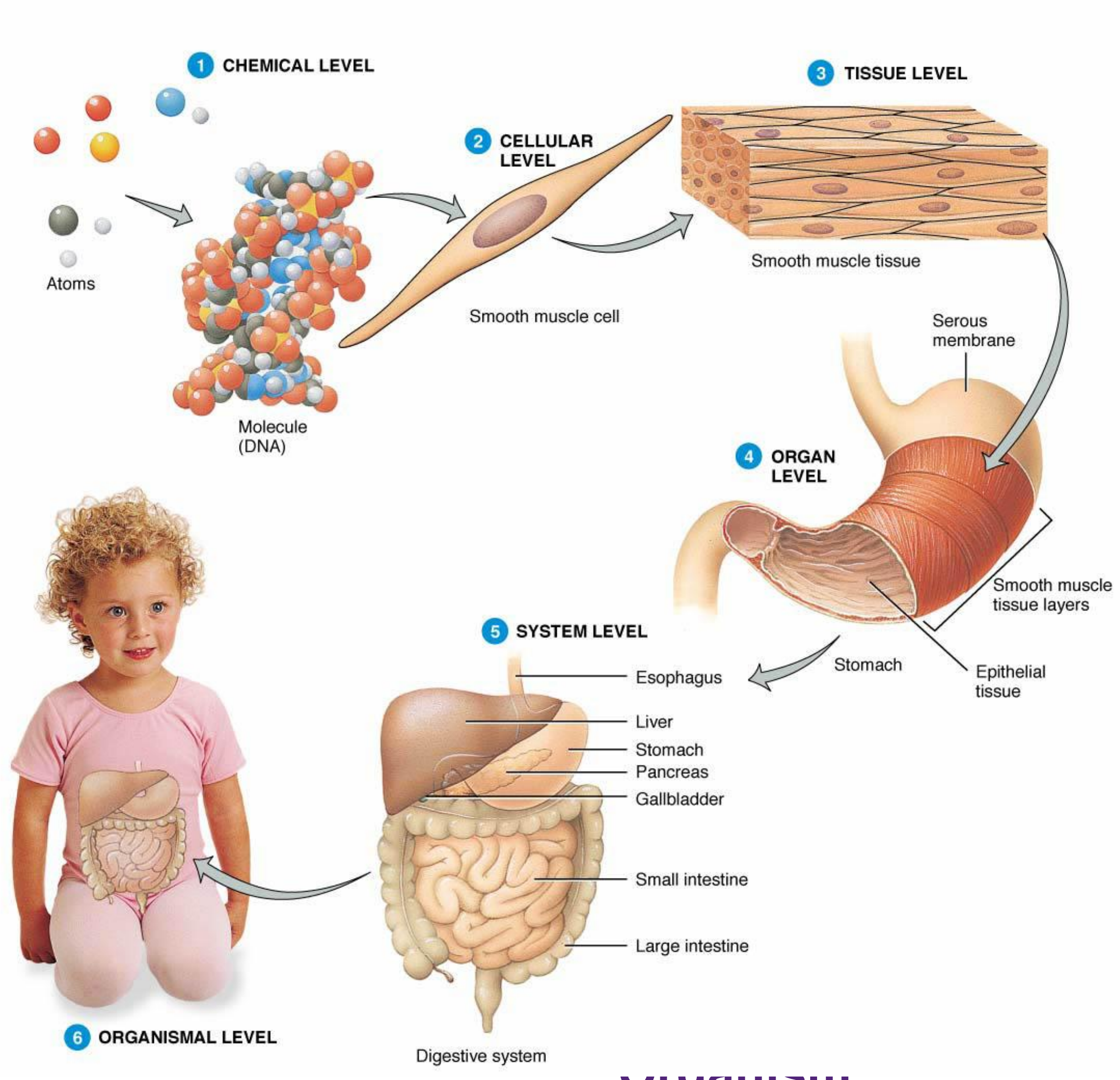
Cellular

Tissue

Organ

System

Organism



Source: http://classroom.sdmesa.edu/eschmid/Chapter1-zoo145.htm

Activity

Latin root	***Meaning***	***Example***
Aden-	*Gland*	*Adenopathy (disease in a gland)*
Angi-		
Arthr-		
Bronch-		
Carcin-		
Cardi-		
Carp-		
Chol-		
Derm-		
Erythro-		
Gastr-		
Hemat-		
Path-		
Sept-		

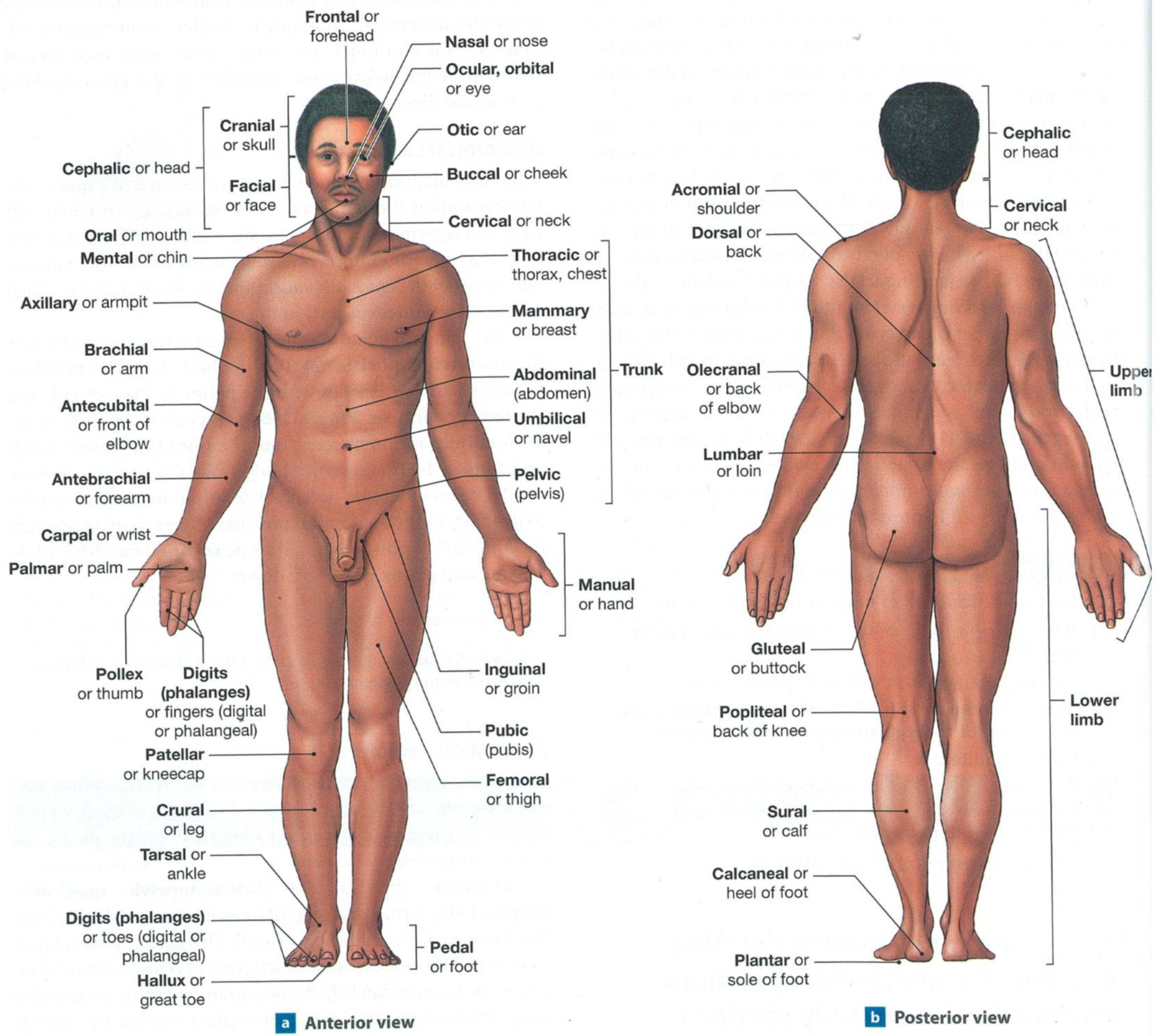
Anatomical Position

- Anatomical position allows us to describe the position of structures in relation to their surroundings e.g. the heart lies above the diaphragm
- Anatomical position avoids confusion as to whether a body is lying down or standing up
- Structures are always described as they are to the subject rather than as they appear to you the observer

* When looking at a person in anatomical position, their right side will be on your left

Anatomical Landmarks

Understanding the terms and their etymology (origins) will help you remember both the location of a particular structure and its name.



Source: Martini, F., Nath, J., & Bartholomew, E. (2012). Fundamentals of Anatomy and Physiology, 9th edition, Pearson International: San Francisco.

Anatomical Position

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Subject stands erect facing the observer

The head is level

The eyes face directly forward

The feet are flat on the floor, directed forward

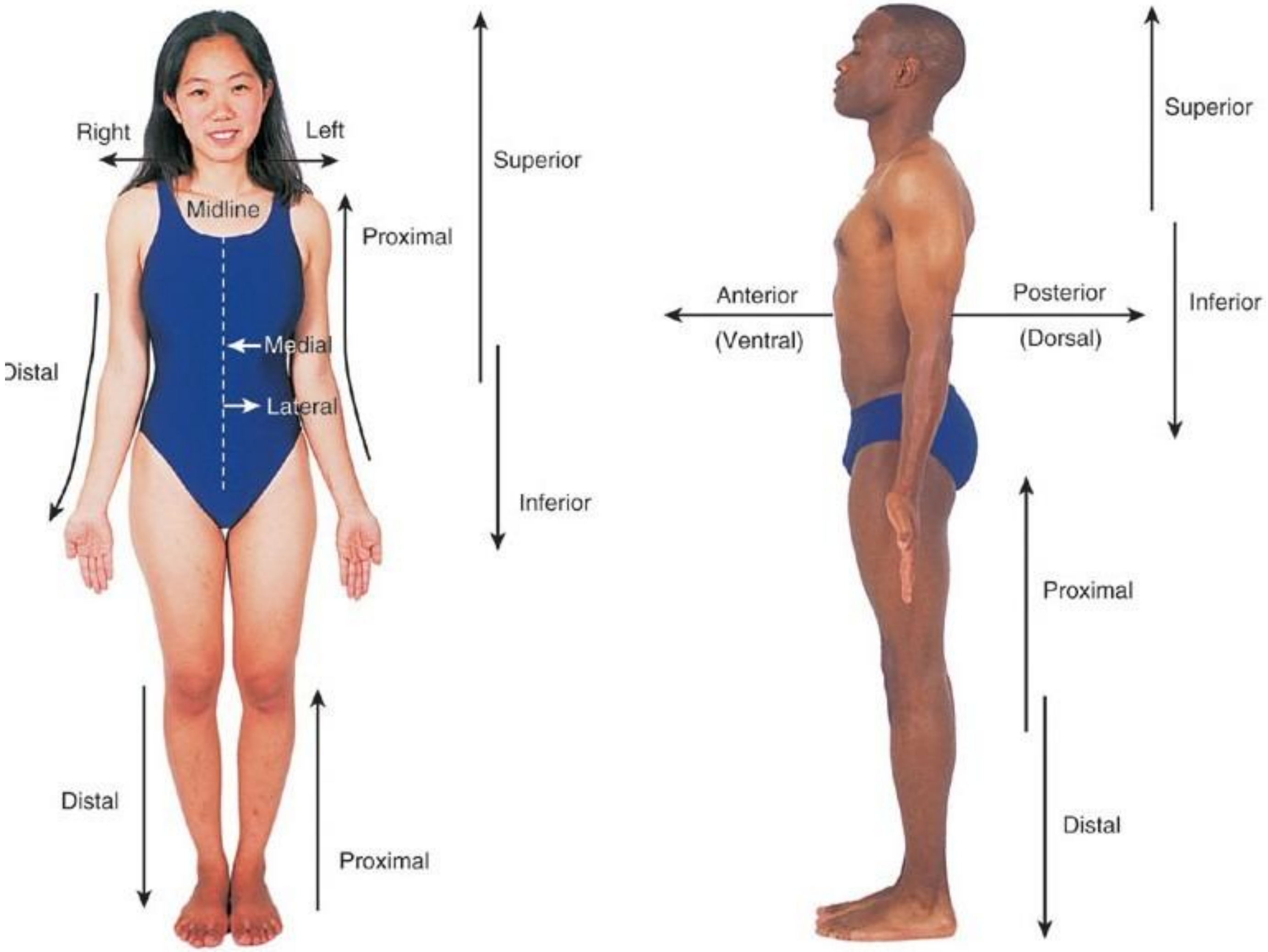
The upper limbs are at the sides with palms turned forward

In anatomical position the body is upright.

When the body is reclined there can be two positions described:

Prone – face down

Supine – face up



Source: http://upanya.blogspot.com.au/2011/06/anatomical-positions-directions-and.html

Activity 2: Getting In Position

Challenge 1:

Place one hand on your clavicle, move it superiorly, posteriorly, inferiorly

Challenge 2:

Put your finger on a spot anterior to your heart, just inferior to your mandible and lateral to your trachea

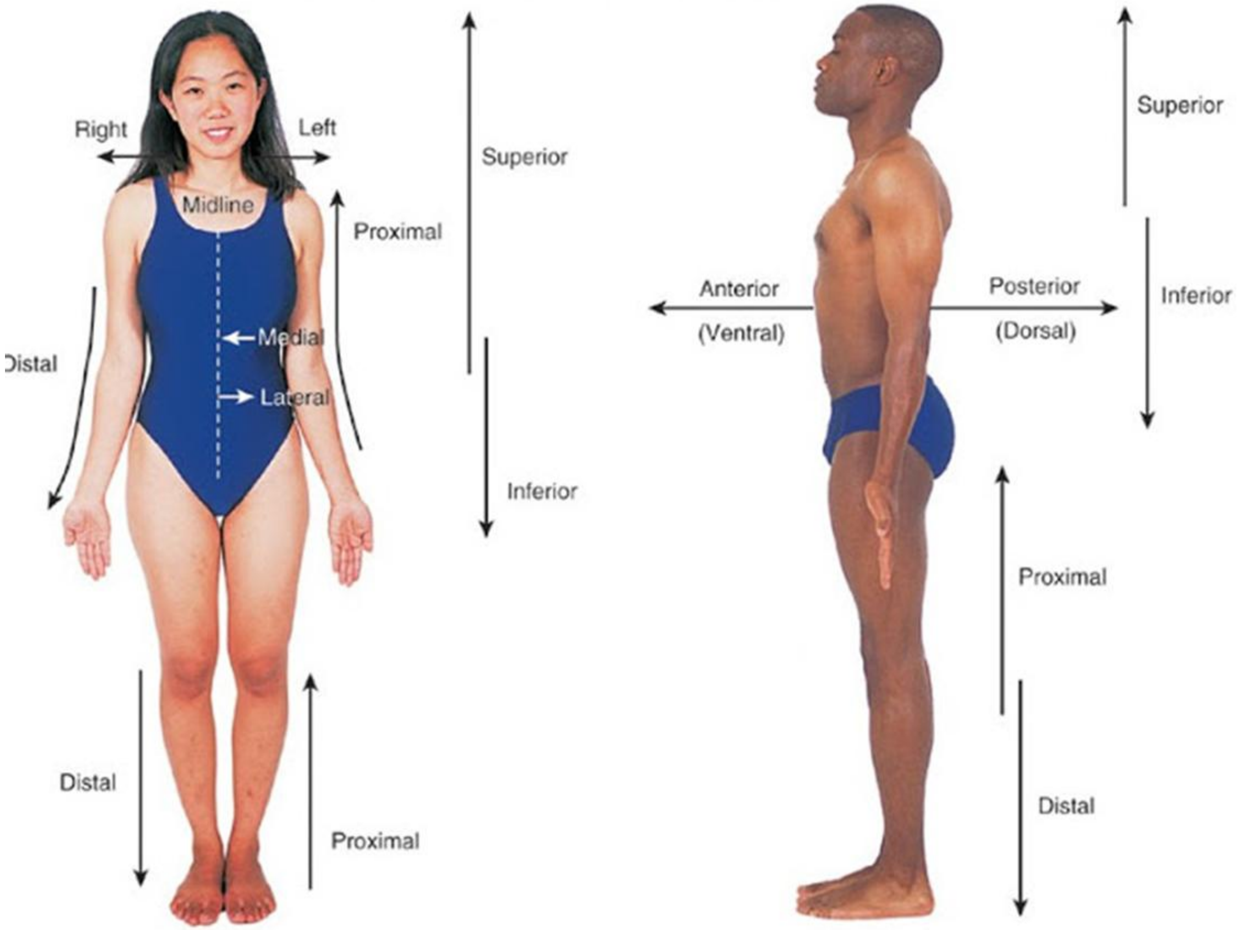
Directional Terms

Directional terms are used to locate various body structures and to describe the position of one body part relative to another. Directional terms are related to standard anatomical position.

Midline is an imaginary vertical line that divides the body into equal right and left sides.

Intermediate is between two structures.

Planes are imaginary flat surfaces that pass through the body parts.



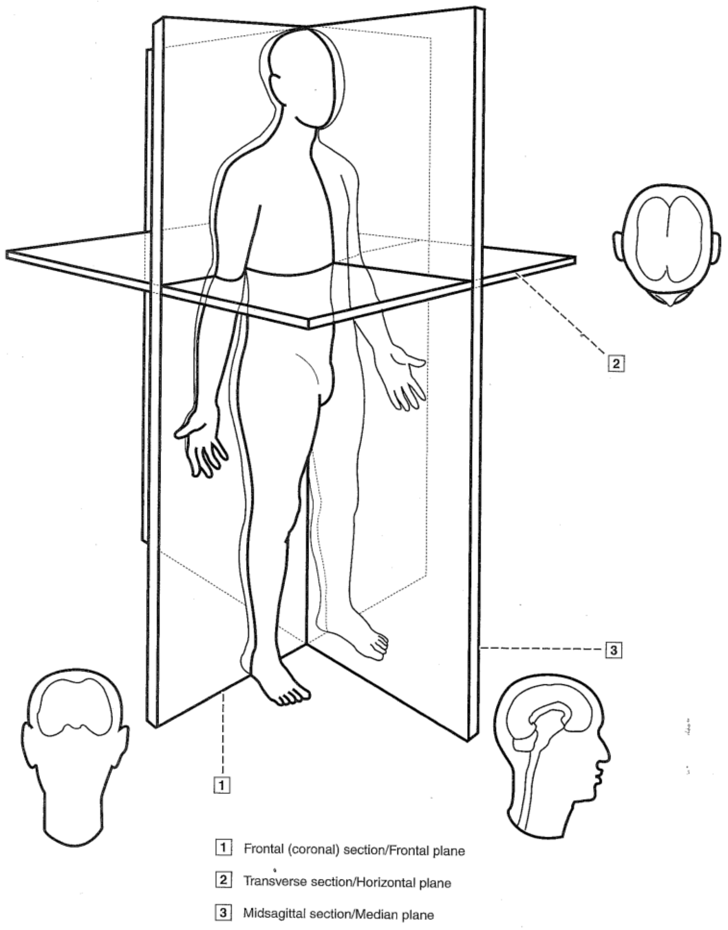
Source: http://upanya.blogspot.com.au/2011/06/anatomical-positions-directions-and.html

Body planes and sections

A **plane** is an axis, an imaginary flat surface, that passes through the body parts.

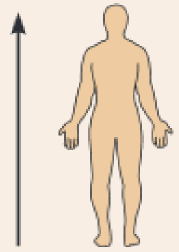
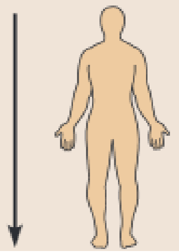
Three planes are necessary to describe any three-dimensional object.

- A **sagittal section** is cut along the lengthwise, or longitudinal, plane of the body
- A **frontal section** is a cut along a lengthwise plane that divides the body into anterior and posterior parts.
- A **transverse section** is a cut along a horizontal plane, dividing the body or organ into superior and inferior parts.



Directional Terms: Superior & Inferior

Superior (cranial) and inferior (caudal)

Superior (cranial)	Toward the head end or upper part of a structure or the body; above	
Inferior (caudal)	Away from the head end or toward the lower part of a structure or the body; below	

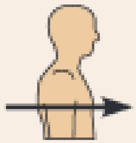
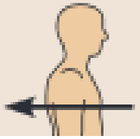
Examples:

- The knees are ___________________ to the hips.
- In humans, the cranial border of the pelvis is __________________ to the thigh.

Picture source: Marieb, E. & Hoehn, K. (2007). Human Anatomy & Physiology, 7th Edition. Pearson: San Francisco, CA.

Directional Terms: Anterior & Posterior

Anterior (ventral) and posterior (dorsal)

Anterior (ventral)*	Toward or at the front of the body; in front of	
Posterior (dorsal)*	Toward or at the back of the body; behind	

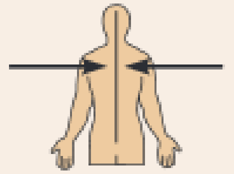
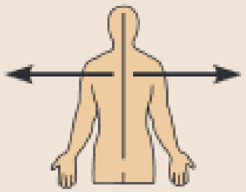
Examples:

- The Sternum (breastbone) is _________________ to the spine
- The heart is ________________ to the breastbone

Picture source: Marieb, E. & Hoehn, K. (2007). Human Anatomy & Physiology, 7th Edition. Pearson: San Francisco, CA.

Directional Terms: Medial & Lateral

Direction with reference to the midline of the body (median plane)

Medial	Toward or at the midline of the body; on the inner side of	
Lateral	Away from the midline of the body; on the outer side of	

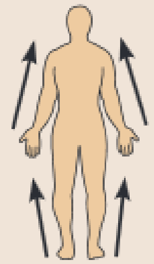
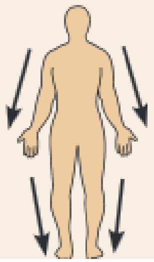
Examples:

- The heart is ______________ to the arm
- The arms are _______________ to the chest

Picture source: Marieb, E. & Hoehn, K. (2007). Human Anatomy & Physiology, 7th Edition. Pearson: San Francisco, CA.

Directional Terms: Proximal & Distal

Used in reference to relative locations of parts on the limbs

Proximal	Closer to the origin of the body part or the point of attachment of a limb to the body trunk	
Distal	Farther from the origin of a body part or the point of attachment of a limb to the body trunk	

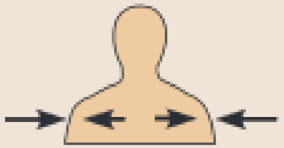
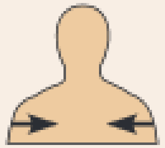
Examples:

- The thigh is _______________ to the foot
- Moving ______________ from the elbow brings you to the wrist

Picture source: Marieb, E. & Hoehn, K. (2007). Human Anatomy & Physiology, 7th Edition. Pearson: San Francisco, CA.

Directional Terms: Superficial & Deep

Superficial (external) and deep (internal)

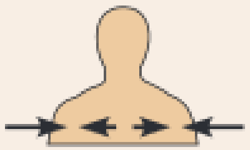
Superficial (external)	Toward or at the body surface	
Deep (internal)	Away from the body surface; more internal	

Examples:

- The bone of the thigh is ______________ to the surrounding skeletal muscles
- The skin is ___________________ to the skeletal muscles

Picture source: Marieb, E. & Hoehn, K. (2007). Human Anatomy & Physiology, 7th Edition. Pearson: San Francisco, CA.

Directional Terms: Intermediate

Intermediate	Between a more medial and a more lateral structure	

Examples:

- The collarbone is ______________ between the Sternum (breastbone) and shoulder

Picture source: Marieb, E. & Hoehn, K. (2007). Human Anatomy & Physiology, 7th Edition. Pearson: San Francisco, CA.

Activity 3

1. The head is ______________ to the abdomen

2. The navel is on the ________________ surface of the trunk

3. Moving ___________ from the arm across the chest surface brings you to the sternum

4. The navel is _______________ to the chin

5. The skin is _______________ to the underlying structures

6. The shoulder(Scapula) blade is located _____________ to the rib cage

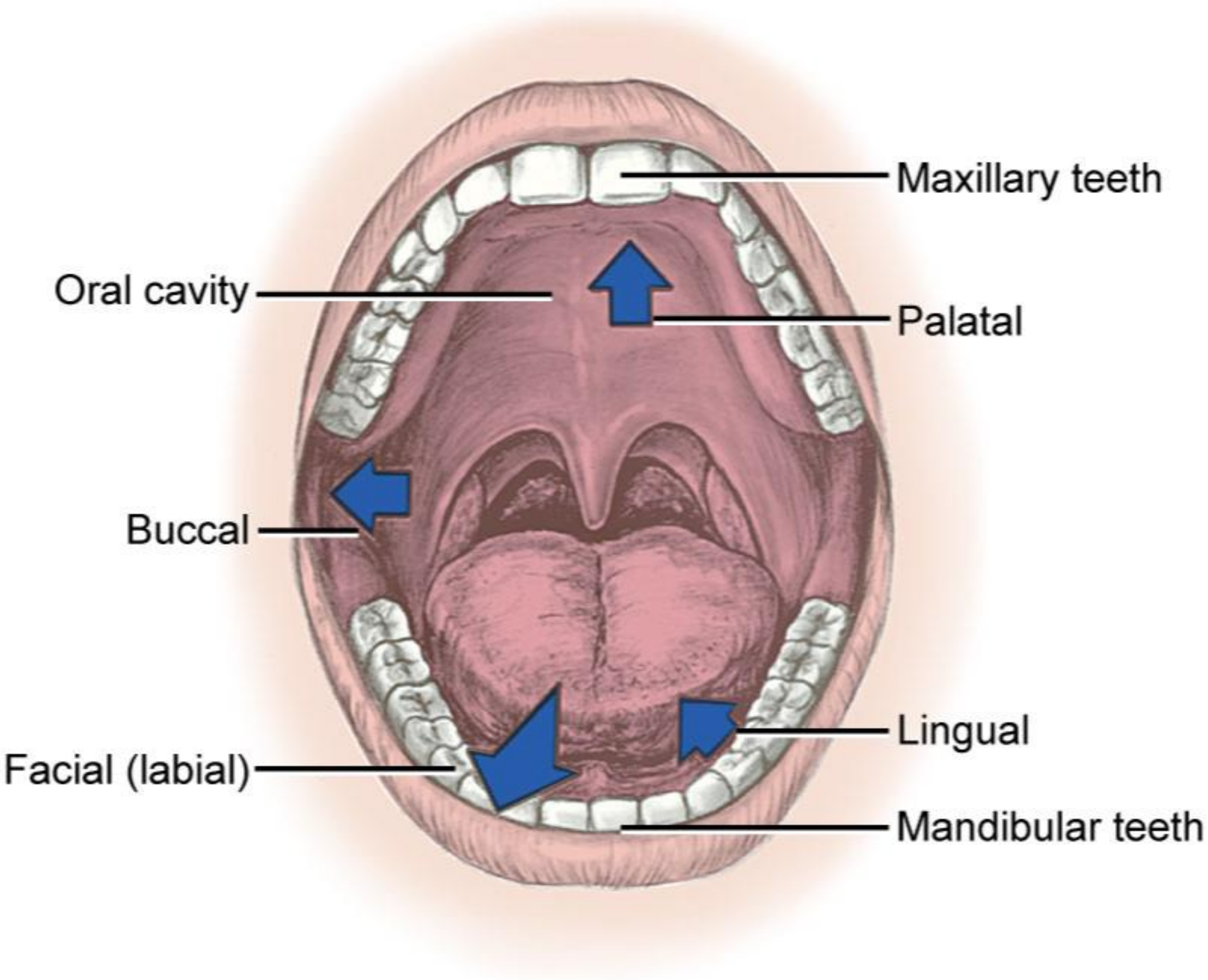
7. Moving _______________ from the wrist brings you to the elbow

8. Moving _______________ from the nose brings you to the cheeks

9. The fingers are ______________ to the wrist

10. The lungs are _______________ to the skin

The Oral Cavity



From Fehrenbach MJ, Herring SW: *Illustrated anatomy of the head and neck*, ed 4, St Louis, 2012, Saunders/Elsevier.

Fig. 2-1. Oral cavity and the jaws with the designation of the orientational terms *(arrows) facial, labial, buccal, palatal,* and *lingual.*

Activity 4: Take-home question The oral cavity

1. The lips of the face are the ________________ border of the oral cavity.

2. The pharynx or throat is the _____________ boundary of the oral cavity.

3. The cheeks of the face are the ____________ boundaries of the oral cavity.

4. The palate is the ____________ boundary of the oral cavity.

5. The floor of the mouth is the _____________ boundary of the oral cavity.

Activity 5: Getting In Position

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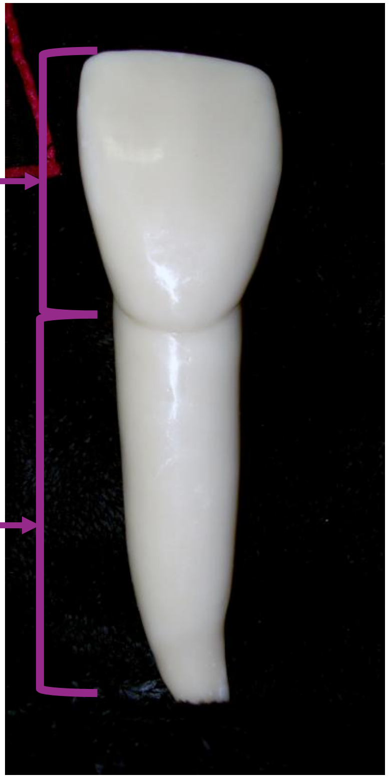
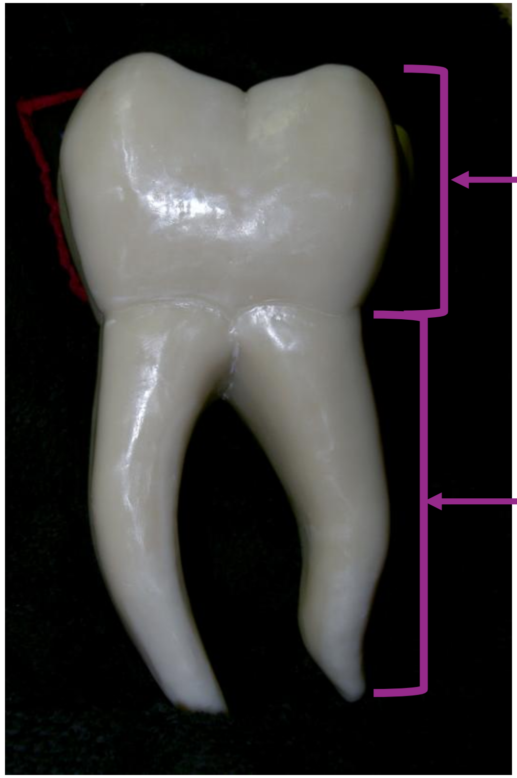
Describe the position of the red dot in these patients.

Anatomical crown

Anatomical root

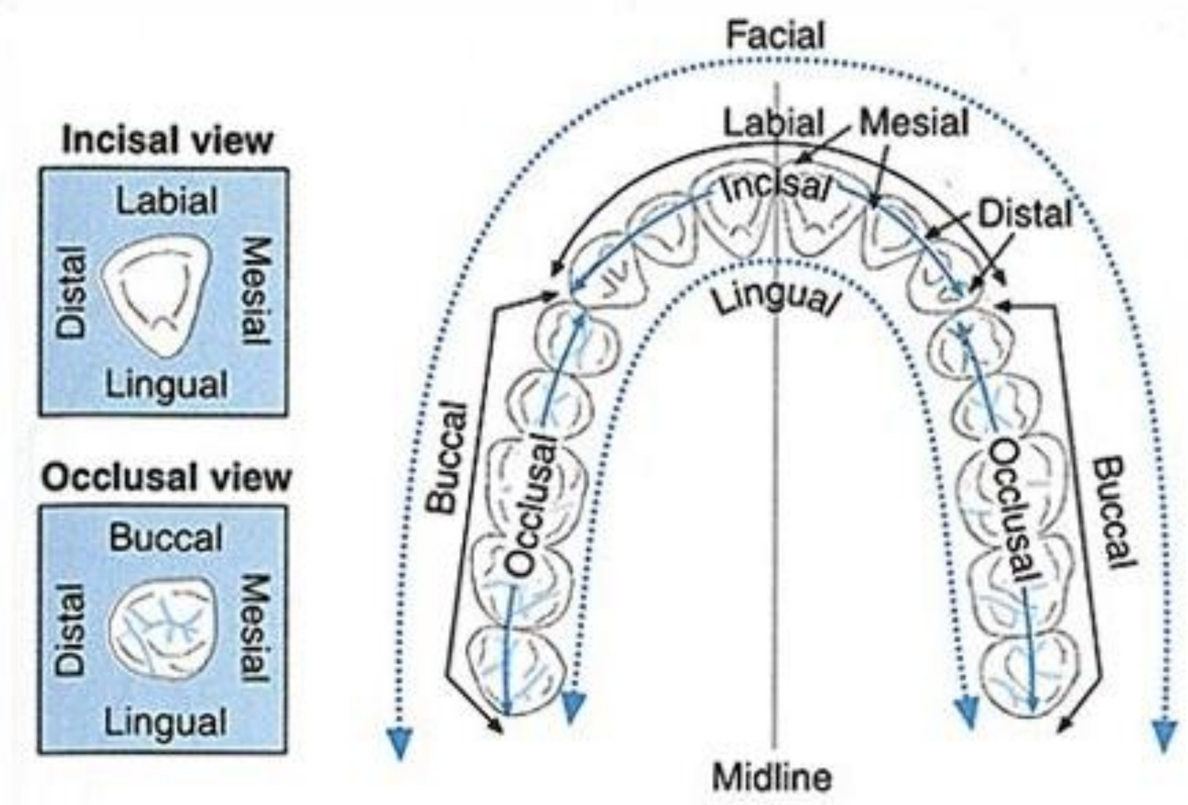


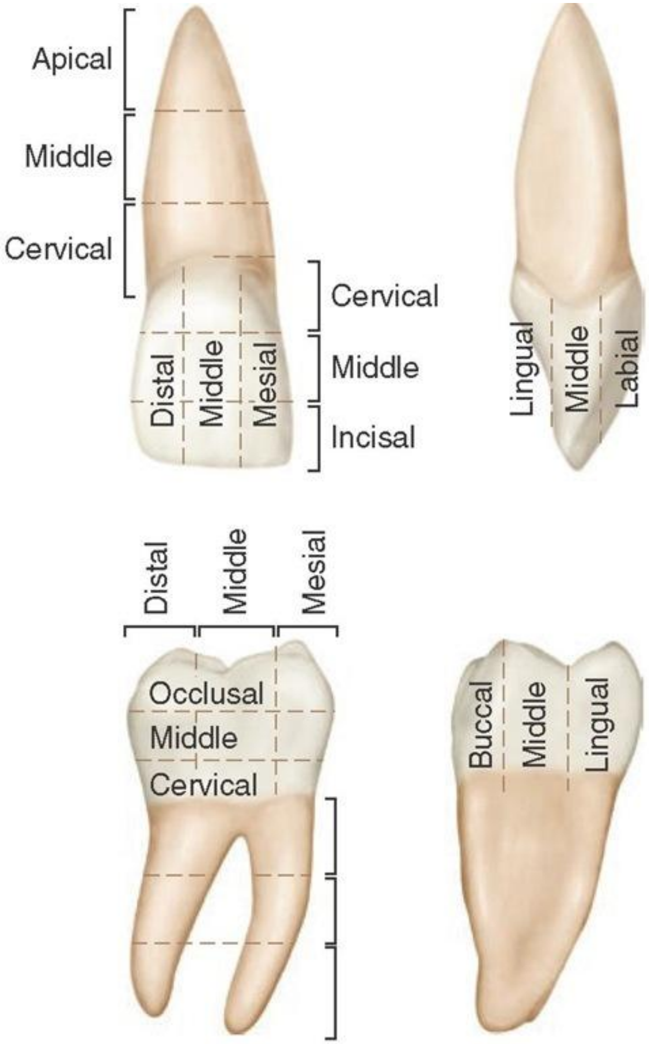


Anatomical crown

Anatomical root

Source: Dr. Sepanta Hosseinpour





Source: http://classroom.sdmesa.edu/eschmid/Chapter1-zoo145.htm

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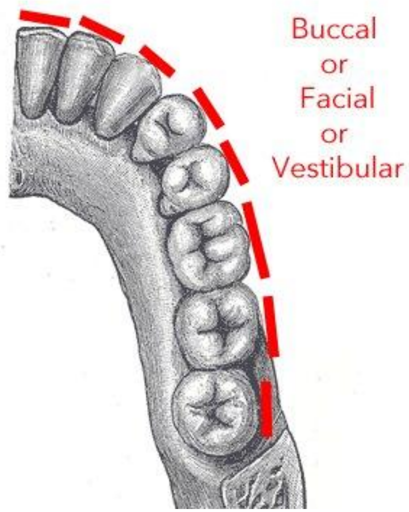
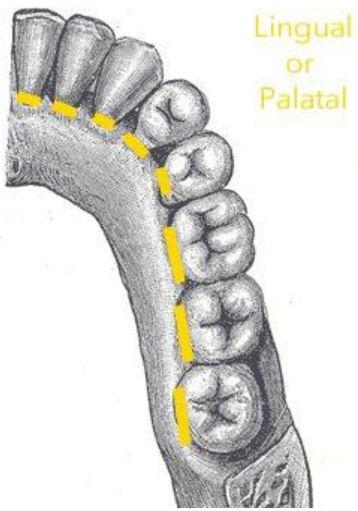
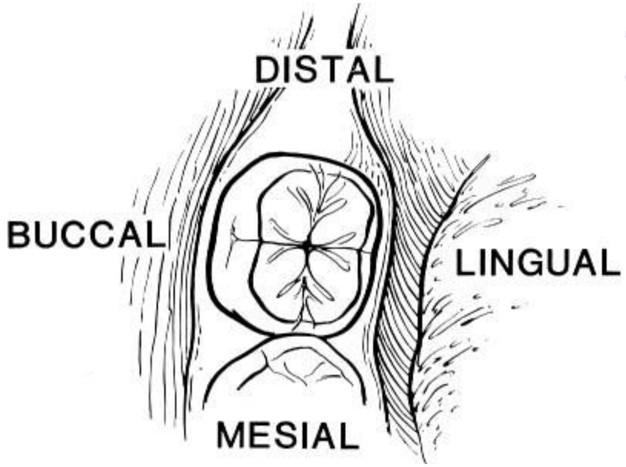
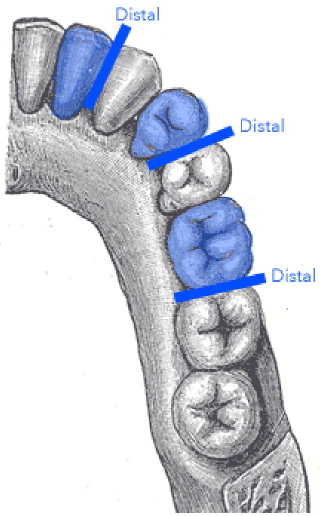


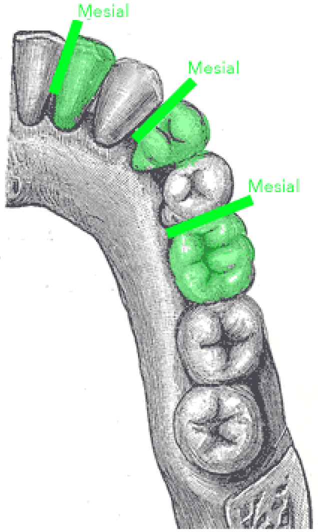






Distal of the highlighted teeth

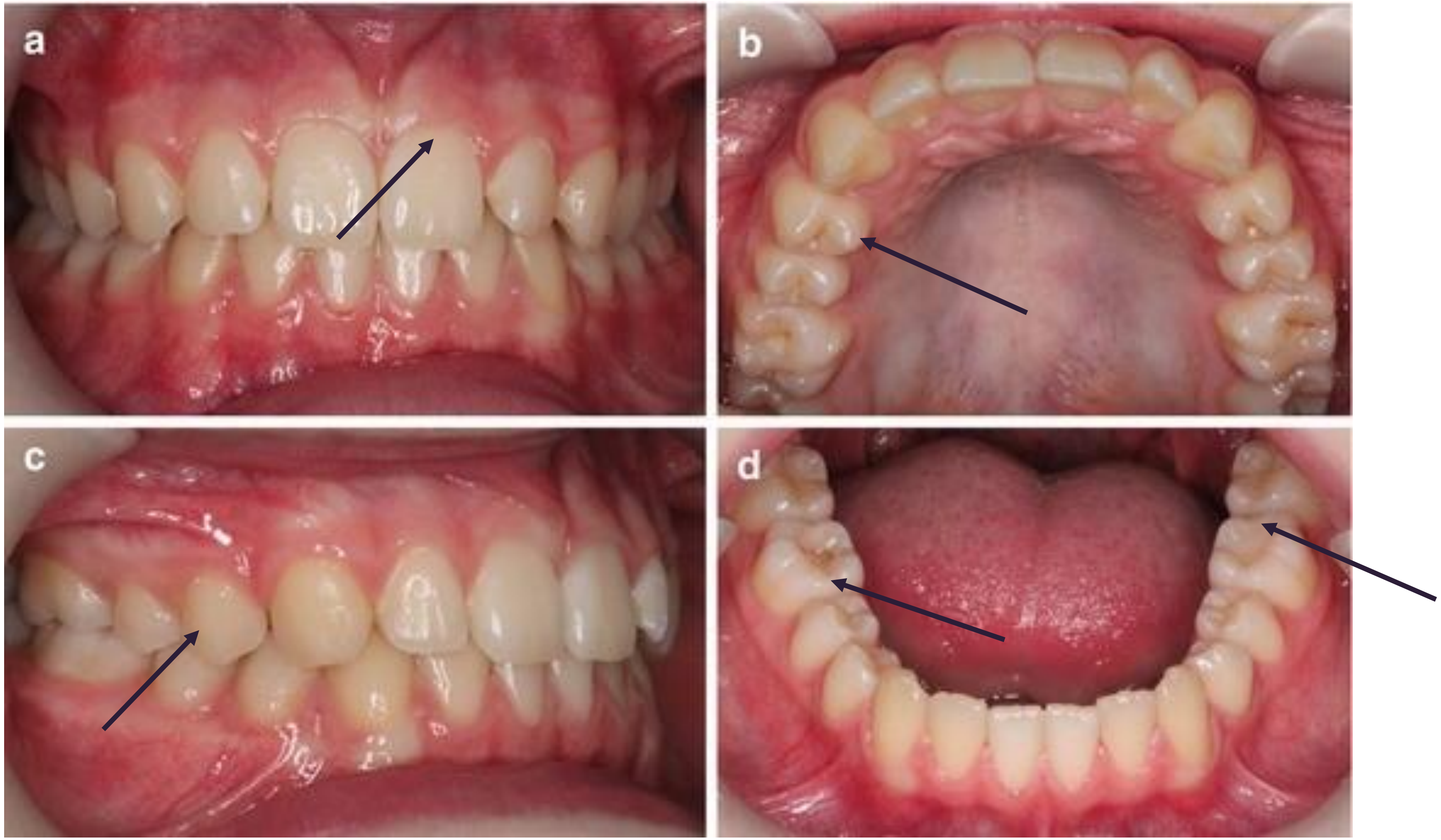
The distal surface is the surface of the tooth that is further from the middle or front of the jaw



Mesial of the highlighted teeth

The mesial surface is the surface of the tooth that is closer to the middle or front of the jaw

Which part of the tooth is indicated by black arrow?



Summary

Anatomy is the study of structure.

Physiology is the study of function.

The correlation between structure and function is a core theme of biology (the study of life)

Everything in your body is organised via a hierarchy: Organism > System > Organ > Tissue > Cellular > Chemical

Anatomical position allows us to describe the position of structures in relation to their surroundings

Directional terms are used to locate various body structures and to describe the position of one body part relative to another.

Directional terms are related to standard anatomical position.

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Thank you

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Slides adapted from
Drs Bartle, Hosseinpour and Fernandez (2018-2025)